

Characterization of low temperature reactions in the standard Cooperative Fuel Research (CFR) Engine

Abstract

Up to date many proposals for the fuel rating in the Spark Ignition (SI) engine have been suggested and there is still no consensus on this and the industry is still using RON and MON tests to rate the fuels and there is a need to come up with new fuel rating system. The knocking tendency of the fuel in SI engines is primarily governed by the end-gas auto-ignition. Another combustion mode, Homogeneous Charge Compression Ignition (HCCI) is also driven by auto-ignition of the complete charge inside the cylinder. Fundamentally, the combustion process in both combustion modes is driven by auto-ignition and HCCI combustion mode can be used to understand the knocking behavior in SI engines. The lean combustion environment in HCCI mode provides a good platform to replicate the operating conditions of modern SI engines which are operating at boosted pressures and low intake temperatures and understanding of the fuel knocking behavior under such conditions is vital for achieving high efficient engines. Therefore in this study, HCCI combustion will be used in the standard CFR engine to understand the auto-ignition behavior of the fuels for SI engines. For this purpose, three fuel blends were selected which had the research octane number equal to 90. The standard CFR engine was operated with varying intake pressures and temperatures under HCCI combustion mode. The Lund-Chevron HCCI fuel number was used to rate the fuels and this was compared with RON and MON of the blends. It was found that HCCI combustion could be used to rate the fuels with the standard CFR engine with minor modifications to accommodate the boosted conditions without affecting the geometry and the flow inside the CFR engine. Low temperature reactions were observed and were correlated with the Lund-Chevron HCCI fuel numbers.

Introduction

The anti-knock quality of the fuel is described historically using the two parameters: Research Octane Number (RON) and Motor Octane Number (MON) in the Spark Ignition engines. The testing of these two octane numbers is carried out in the Cooperative Fuel Research (CFR) engine according to the standard ASTM methods [1, 2]. The scale of the RON and MON is based on two paraffinic fuels, iso-octane with the value of 100 and n-heptane with a value of 0 [3, 4]. The mixture of these two fuels is referred to as Primary Reference Fuels (PRFs). The operating conditions for RON and MON are shown in Table 1. The main difference between the two tests are the engine speed and intake temperature.

Table 1 : Operating conditions of the engine for RON and MON [1, 2]

Engine parameters	Research Octane Number (RON)	Motor Octane Number (MON)
Engine speed (rpm)	600	900
Inlet Air Temperature (°C)	52	149
Spark Timing (BTDC)	13	14-26
Compression ratio	4-18	4-18

The RON and MON test methods were established to characterize the knock propensity of the fuels in SI engines at the time. It has since been established that both RON and MON do not fully describe the anti-knock quality of practical fuels [5-7]. Kalghatgi proposed a practical method to correlate the RON and MON with the engine operating conditions by proposing the Octane Index (OI) to describe the anti-knock quality of the real-world fuels in modern SI engines [3]. OI is a linear function of both RON and Sensitivity (S), where S is given by RON-MON and OI is given [3] by :

$$OI = RON - K * S \quad (1)$$

The constant K is depending on the pressure-temperature history of the unburned gas. For the RON test, K is defined to be 0 and for MON test, K is 1 [5, 8].

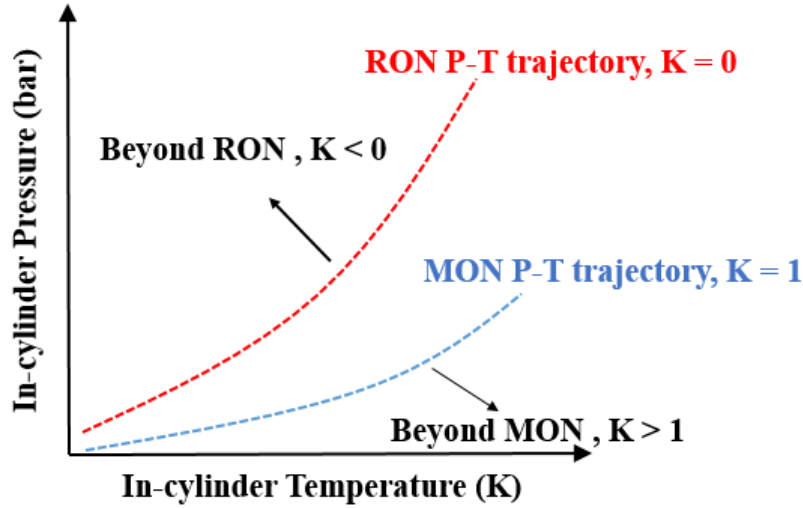


Figure 1 : Cylinder pressure and temperature of the RON and MON compression trajectories [9, 10]

Figure 1 shows the two trajectories of RON and MON and the operating condition beyond RON and MON. The efficiency of future SI engines is limited due to knock [5, 11]. Mainstream technologies such as intercooling and direct fuel injection have been introduced and with that modern SI engines tend to operate at lower cylinder temperatures and higher intake manifold pressures which lead to operation in the beyond RON regime where $K < 0$ as shown in Figure 1 [7, 8, 12].

Furthermore, according to Wang et al. the physiochemical properties such as ignition delay, octane sensitivity and boiling range of commercial gasoline and biofuels differ significantly from PRF fuels (iso-octane and n-heptane) [12]. Biofuels such as ethanol have a RON equal to 107 [4] and in order to obtain the octane number beyond the paraffinic fuel scale, tetraethyl lead is added to iso-octane to define a new rating scale beyond 100 [1]. Swarts et al also highlighted the inconsistencies between the understanding of the auto-ignition and the knock intensity measured from the knockmeter suggesting that the octane data obtained from the CFR engine should be used with care when trying to interpret the fuel's knocking behavior [13, 14]. And it has therefore been argued by Wang et al. that the hardware and fueling method in modern engines and that of the standard CFR engine differ significantly which again questions the applicability of RON and MON [12].

Knock in SI engines is caused by auto-ignition of the end gas [8, 9, 11, 15]. A fraction of the total premixed air and fuel charge is auto-ignited and the chemical energy is converted to heat energy causing a rapid increase in the pressure leading to knock. It is quite difficult to distinguish the low temperature reactions which are hidden behind the extensive heat release from the propagating flame. Another combustion mode which has gained popularity among the engine research community due to ultra-low NO_x and PM emissions is Homogeneous Charge Compression Ignition (HCCI) in which the auto-ignition occurs for the whole charge in the cylinder [16, 17]. Fundamentally the combustion in both knocking SI and HCCI combustion mode is driven by auto-ignition [18]. It has also been found that by studying HCCI combustion, auto-ignition can be understood for the range of pressure-temperature conditions which are not possible in SI engine due to knock [19].

The HCCI combustion is driven by chemical kinetics and is dependent on temperature, fuel chemistry and the time available for the chemical reactions [20]. The low temperature heat release (LTHR) has an important role in the HCCI combustion and its occurrence and magnitude can vary depending on the fuel type [21]. For paraffins, the magnitude of LTHR is higher compared to olefins [21]. Najt et al experimentally and numerically investigated the basic mechanism behind the HCCI combustion with a focus on how ignition and energy release processes are controlled by the various operating parameters such as compression ratio, equivalence ratio, engine speed, fuel type and etc. [22]. For the experimental study, the CFR engine with a pancake combustion chamber and shrouded intake valve was used [22]. It was found that the HCCI combustion was driven purely by chemical kinetics and the ignition process is controlled by the low temperature oxidation reactions whereas the energy release process is controlled by high temperature hydrocarbon oxidation kinetics [22]. Liu et al experimentally investigated the effect of fuel composition and operating conditions using the modified four-cylinder direct-injection diesel engine [23]. Six fuels were tested which included PRFs, mixtures of ethanol/PRFs and two unleaded commercial gasoline fuels [23]. The RON of all the fuels was kept at 90 [23]. It was found that the effect of fuel properties on HCCI combustion depended on the operating conditions [23]. It was concluded that the fuels with higher sensitivities ($S = \text{RON} - \text{MON}$) will likely help to extend the operating range of HCCI combustion provided better control strategies such as intake pressure, temperature and etc. are used [23]. A HCCI study by Shibata et al, investigated the effect of fuel properties on low (LTHR) and high temperature heat release (HTHR). Four fuels with different octane numbers were experimentally tested in the supercharged 4-cylinder engine [21]. The four fuels consisted of PRF 70, 80, 92 and a regular gasoline with RON 90. It was found that the amount and the start of LTHR was depended on the octane number of the fuel [21]. For low octane fuel such as PRF 70, the LTHR was maximum. The combustion phasing and the amount of LTHR had a direct link with the phasing of the HTHR [21]. Hosseini et al experimentally tested the effect of engine speed on HCCI combustion with diesel like fuels, n-heptane (having a two-stage ignition), and iso-octane (having a single-stage combustion) in the CFR engine [20]. The investigated speed range was from 600 rpm to 1700 rpm and the fuel was injected by using the port fuel injection [20]. As the engine speed was increased, the LTHR was reduced which retarded the HTHR timing [20]. In order to keep the same combustion timing, the two controlling parameters, intake temperature and boost pressure were used [20]. It was found that for single-stage combustion fuels such as iso-octane, the intake boosting or heating was small but for dual-stage combustion fuels, a significant amount of variation was required [20].

This paper is written with the intention of explaining how the analysis of HCCI operation in the standard CFR engine can help to detect low temperature reactions which could provide vital information towards understanding knocking combustion behavior in SI engines. For this purpose, the CFR engine was run with a range of inlet pressures and temperatures enabling operation in the “beyond RON and MON regime” associated with operating conditions relevant for modern engines. Elsewhere, it has already been shown by Trudesson et al. that the Lund Chevron-HCCI fuel number is a more useful parameter than RON and MON to describe a fuel’s auto-ignition properties in HCCI combustion mode [24]. The Lund Chevron-HCCI fuel number test protocol was therefore utilized as a means of rating three fuel blends in this study. Detailed heat release analyses were performed to analyze the manifestation of low temperature reactions. The most unique thing about this HCCI study is that it was performed in an unaltered CFR engine with the carburetors. All previous studies in the literature were done using the modified CFR engine with t port fuel injection.

Test Fuels and operating conditions

Three fuel blends were selected for this study. N-heptane was selected as a base fuel and iso-octane, toluene and ethanol were added as octane boosters to reach an RON of 90 in all three cases. Table 2 illustrates the volume percentages of iso-octane, toluene and ethanol added to n-heptane and the operating conditions for HCCI testing. In addition to HCCI, SI testing was also conducted

to obtain Research Octane Number. All these parameters will be discussed in details in the octane numbers evaluation section. An intake pressure sweep was conducted for all five intake air/fuel temperatures.

Table 2 : Fuel blends and the operating conditions under which they were tested

	Fuels blended with volume percentage				Operating conditions
	Base fuel	Octane boosters			
Fuel blends	N- Heptane	Ethanol	Toluene	Iso-octane	Inlet pressure (bar) = 0.9, 1.0, 1.15 and 1.3 Inlet air-fuel temperature (±5 °C) = 30,52,80,120,149,180
PRF 90	10	0	0	90	
Toluene/N-heptane	29.5	0	70.5	0	
Ethanol/N-heptane	45	55	0	0	

Experimental Set up – SI and HCCI combustion

The CFR engine used in this investigation was based on the standard CFR F1 RON rating engine, with only minor modifications. Figure 2 shows the schematic of the standard CFR engine, which was operated at 600 rpm. Instead of being naturally aspirated, the intake air was fed by a filtered and dehumidified screw-type facilities air compressor. The intake pressure was controlled by an Alicat Scientific MCR-500LPM mass flow controller. The thin-walled 6" diameter intake transfer tube used in the standard CFR engine was replaced by a thicker-walled pipe with the same diameter and similar length. To be able to use the standard carburetor and adjustable-height equilibrium fuel bowls, the carburetor breather holes were connected to the 6" diameter transfer tube to equalize pressure. For near stoichiometric operation, the standard carburetor horizontal and vertical fuel jets were used. However, smaller horizontal jets were made for lean (near lambda 3) HCCI operation. The standard MON mixture air temperature heater, mounted downstream of the carburetor, was used to control the temperature of the air and fuel entering the intake port. The standard knock meter was replaced by a flush-mounted Kistler 6045-AU20 pressure transducer to acquire the pressure data. Other measuring instruments used to capture the data is shown in the Table 3.

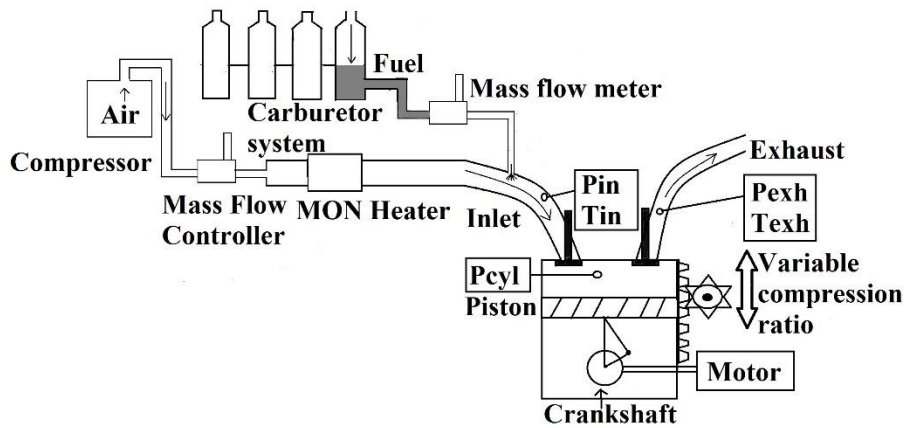


Figure 2 : Schematic of the CFR engine set-up used for this study

Table 3 : CFR measuring instruments [25]

Crankshaft angle-based measurements	
Crank-angle based DAQ	AVL IndiCom & crankshaft encorder
Spark timing	Coil wire current clamp
Intake pressure	Kulite high-speed 2.0 bara pressure transducer
Exhaust pressure	Kulite water-cooled high-speed 3.5 bara pressure transducer
High Frequency Data Acquisition	
DAQ	AVL 620 IndiSet
Cylinder Pressure	Kistler 6045 AU20
Time-based measurements	
Time-based DAQ	LabVIEW
Intake pressure	Setra 3550 pressure transducer
Exhaust pressure	Setra 3550 pressure transducer
Intake, mixture, exhaust, coolant and oil temperatures	K- Type Thermocouples
Fuel rate	Emerson CMF010M
Air rate	Alicat mass flow controller
Lambda Bosch wide-band lambda	Bosch wide-band lambda sensor
ASTM knock units	Data-logged knockmeter signal

Octane Number Evaluations under Stoichiometric and Lean conditions

The standard CFR engine was operated under both stoichiometric and lean conditions to obtain the Auto-ignition Metrix for the rating of the fuels. The Auto-ignition Metrix was classified into octane number (RON) and Lund-Chevron HCCI fuel numbers (HCCI # 1_{ATM}, HCCI # 2_{ATM}, HCCI # 1_{1.3bar}, HCCI # 2_{1.3bar}). The Lund-Chevron HCCI fuel numbers were obtained at an excess air ratio of three. A diluted charge is required in HCCI combustion to keep the pressure rise rate below an acceptable level. Five octane indices for both SI and HCCI tests were obtained.

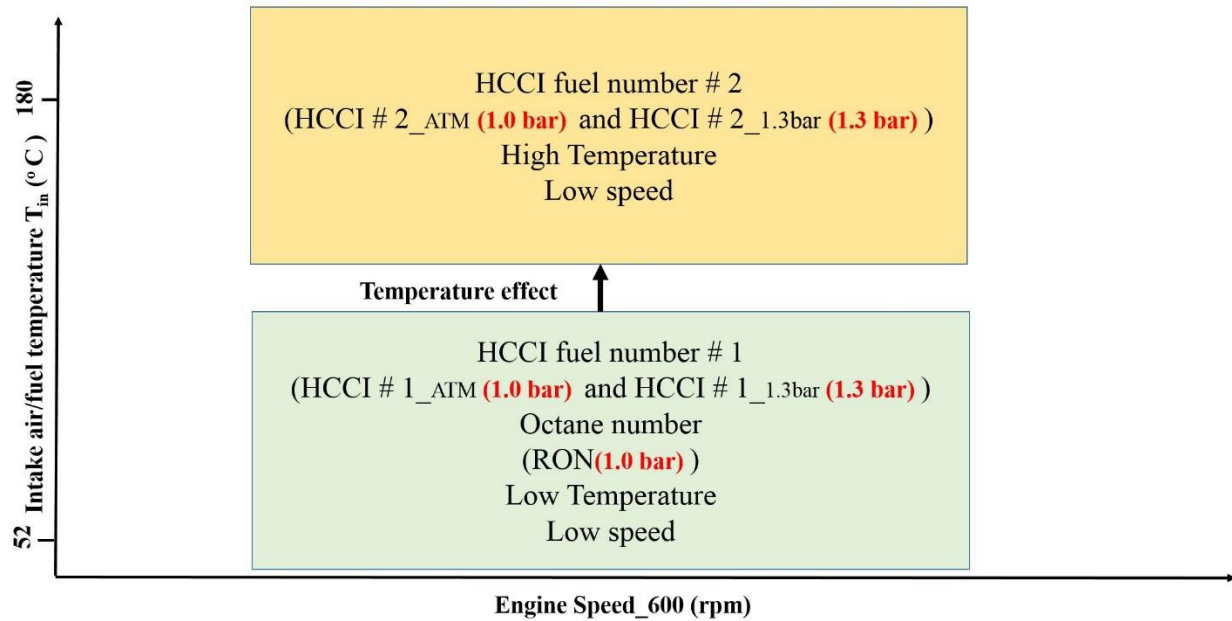


Figure 3 : Operating conditions for octane and HCCI fuel numbers

Figure 3 represents the operating conditions (Inlet air/fuel temperature, Engine speed) under which the Auto-ignition Metrix was obtained. The illustrated RON, HCCI # 1_ATM and HCCI # 2_ATM were obtained by running the engine under naturally aspirated conditions (1.0 bar) and the remaining HCCI fuel numbers, HCCI # 1_1.3bar and HCCI # 2_1.3bar were obtained under boosted (1.3 bar) conditions. For stoichiometric conditions, RON data was obtained using the standard ASTM D2699 protocols [1]. The HCCI fuel numbers were obtained using the transfer function method. The method used to obtain the Lund-Chevron HCCI fuel number requires defining the compression ratios for a given fuel that enables a combustion phasing of CA50 (Crank Angle for 50% heat release) to be maintained at 3° after TDC [24].

A transfer function was used to obtain the Lund-Chevron HCCI fuel numbers by testing PRFs in the range of 70 to 97 research octane number. The transfer function for this study was defined as the conversion procedure of the compression ratio readout to the mixture of iso-octane and n-heptane. Figure 4 shows the obtained transfer function for naturally aspirated and boosted pressure conditions. Two transfer functions, HCCI # 1 (600 rpm, 52°C) and HCCI # 2 (600 rpm, 180°C) both at naturally aspirated and boosted conditions were obtained.

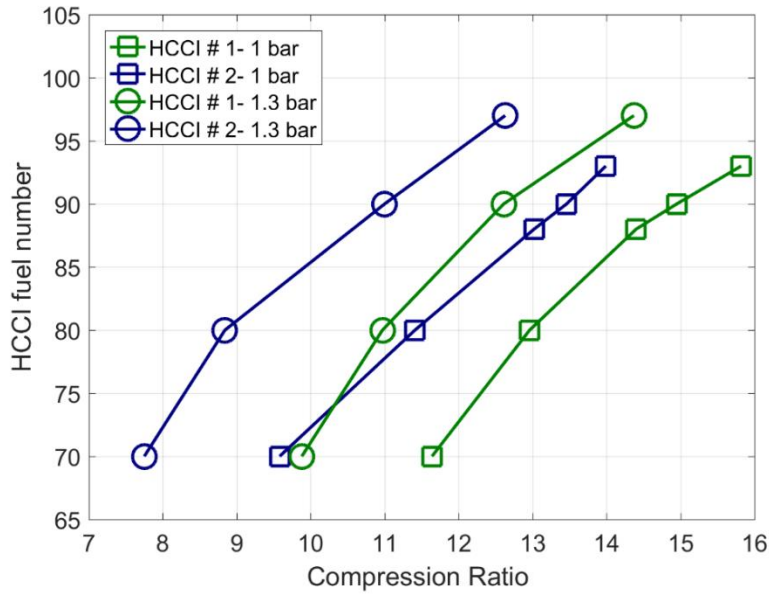


Figure 4: HCCI transfer function

A best fit line was obtained for all four conditions. The best fit line was in the form of a quadratic equation given by:-

$$HCCI \text{ fuel number} = pR_c^2 + qR_c + r \quad (2)$$

Where *HCCI fuel number* represented the percentage of iso-octane by volume in the Primary Reference Fuel mixture and R_c was the compression ratio. The coefficients of the equations were p , q and r . The fuel blends in the Table 2 were tested and the compression ratio was obtained and used along with equation 2 to estimate the HCCI fuel number.

Results

The auto-ignition behavior of the fuel blends was compared in both stoichiometric and lean combustion environment. Figure 5 illustrates the research octane and HCCI fuel numbers for the three fuel blends tested in both HCCI and SI combustion mode. By definition, all the octane and HCCI fuel numbers in all conditions were equal to 90 for PRF 90. The volume percentage of ethanol, iso-octane and toluene into n-heptane were selected to have a Research octane number (RON) approximately equal to 90 and this can be observed through the overlap of the three data points on the far left side of the Figure 5. The MON data was obtained from the **work** done by Trudesson et al [24] and the ASTM MON D2700 protocol [2]. Under lean combustion conditions (HCCI#1_{Atm}), an octane enhancement effect is observed with increased HCCI fuel numbers for toluene/heptane and ethanol/heptane blends when compared to the RON conditions. The octane enhancement was further improved for the boosted conditions (HCCI # 1_{1.3bar}). The opposite effect was found with the high inlet temperature conditions (HCCI#2_{Atm} and HCCI#2_{1.3bar}). The toluene/heptane and ethanol/heptane blends had lower HCCI fuel numbers than PRF 90. This is because lower compression ratios were required for high inlet temperature conditions to have the same combustion phasing of CA50 (Crank Angle for 50% heat release) of 3° after TDC. The boosted HCCI fuel number (HCCI#2_{1.3bar}) was found to be higher at higher inlet air temperature conditions when compared to HCCI#2_{Atm}. Overall, it could be observed that boosted conditions resulted in higher HCCI fuel numbers. When comparing the increased effect of temperature from 52°C to 180°C, for naturally aspirated conditions, the difference in the respective HCCI fuel numbers (HCCI#1_{Atm} and HCCI#2_{Atm}) for the fuel blends was greater when compared with the difference

of the boosted HCCI fuel numbers (HCCI#1_{1.3bar} and HCCI#2_{1.3bar}). The fuel sensitivity ($S = \text{RON} - \text{MON}$) under stoichiometric conditions for ethanol/heptane and toluene/heptane was 9.3 and 11.1 respectively whereas under HCCI conditions, ethanol/heptane has a fuel sensitivity ($S = \text{HCCI\#1}_{\text{Atm}} - \text{HCCI\#2}_{\text{Atm}}$) of 8.2 which is higher than the fuel sensitivity of the toluene/heptane ($S = 5.5$) blend. The opposite effect for fuel sensitivities under the two combustion modes could be due to both the operating conditions (engine speed, intake temperature) and from the effect of lambda.

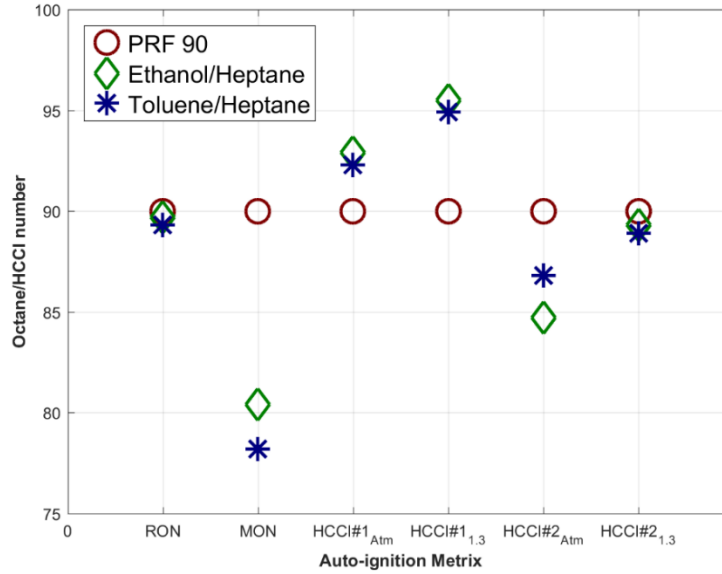


Figure 5 : Octane and HCCI fuel numbers

Two important parameters in the control of HCCI combustion are compression ratio and intake temperature. In order to explore the variation in the compression ratios with the operating conditions (intake pressure/temperature), Figure 6, 7 and 8 illustrates the intake pressure-temperature conditions at which the three fuel blends were tested in HCCI mode. The data points correspond to the compression ratios needed to maintain a constant combustion phasing of CA50 at 3° after TDC. For all three blends, lower compression ratios were required when increasing the inlet pressure to 1.3 bar. The changes in the compression ratio between the data points was smaller at low intake temperatures when compared with high temperatures. This was found to be true for all the three blends. For a given pressure, higher compression ratios were observed at lower inlet air-fuel temperatures as expected. The slopes for PRF 90 were almost parallel to each other whereas for the toluene/heptane blend, the slopes converge at low temperatures and the ethanol/heptane blend appears to diverge from 120°C (shown by a horizontal line on the plot). Overall the conclusion is that the three fuels do not behave the same even though they have the same octane rating (RON). In addition to the above observations, it was realized during the testing that small changes in the compression ratio can lead to large changes in the slopes for Figure 6-8. Therefore, a daily test was carried out on the PRF 90 blend at the start and the end of the experiments to check the consistency of the obtained compression ratios. On average, the uncertainty percentage for PRF 90 was found to be 2%. Figure 6-8 also include the uncertainty bars for the plotted data based on the daily test of PRF 90. These fall within the marker size.

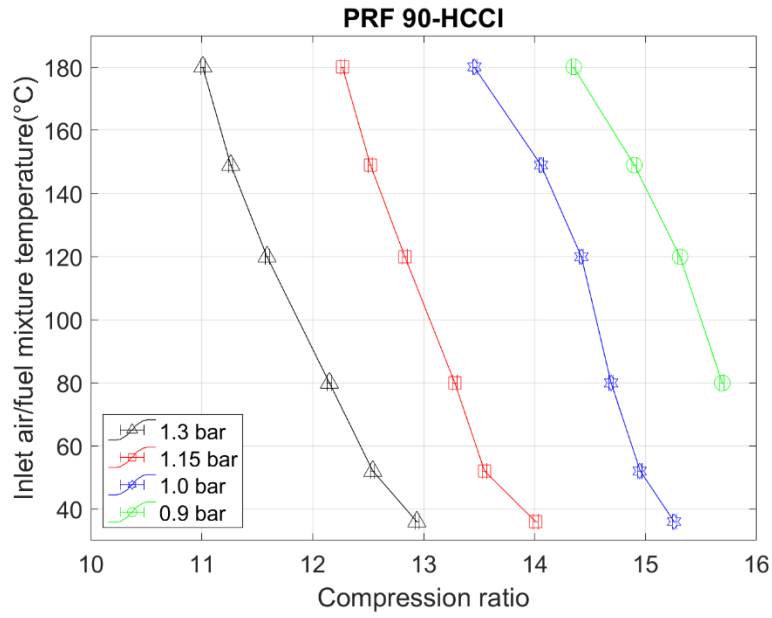


Figure 6: Combinations of intake temperature and compression ratio giving CA50 of 3° after TDC for PRF 90 at four intake pressures

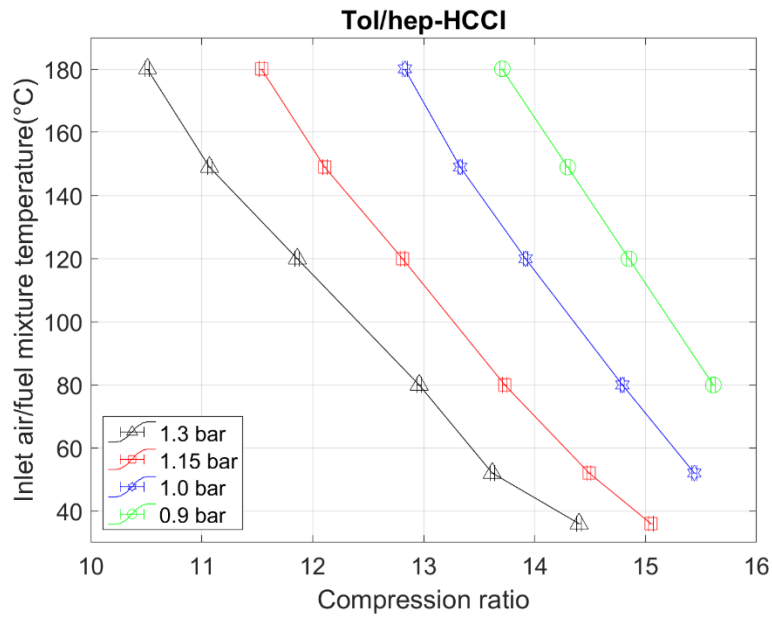


Figure 7 : Combinations of intake temperature and compression ratio giving CA50 of 3° after TDC for the toluene/heptane blend at four intake pressures

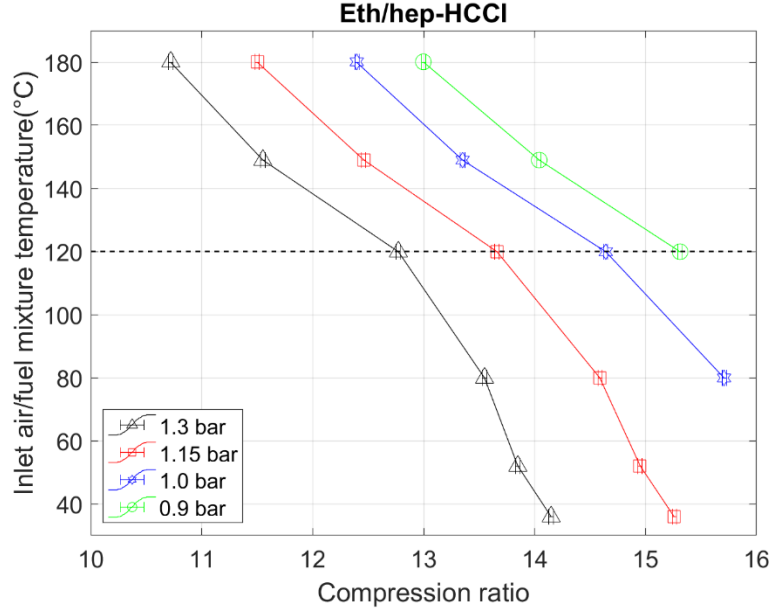


Figure 8: Combinations of intake temperature and compression ratio giving CA50 of 3° after TDC for the ethanol/heptane blend at four intake pressures

Heat release analysis

The heat release analysis reported here is based on the cylinder pressure information extracted from the Kistler cylinder pressure transducer. For each of the fuel blends tested, 300 pressure cycles were recorded. The heat release analysis was based on individual cycles and the reported figures are based on the averaged heat release over the 300 cycles. First law of thermodynamic were used to derive the rate of heat release which was expressed as a function of in-cylinder pressure, volume and ratio of specific heat (γ). Details about the heat release analysis are shown in the Appendix A. The cylinder temperature was estimated by the ideal gas law and the intake valve closure (IVC) conditions. For the data analysis shown here, the temperature at IVC was estimated using the residual gas model described by Heywood [26] and the simplified IVC temperature equation based on compression ratio, intake and exhaust temperature derived by Nikkar et al [27]. The uncertainty in IVC temperature was in an order of 3-5°C. This was based on the uncertainty range defined for the intake and exhaust temperature thermocouples.

The hydrocarbon oxidation is divided into three temperature regimes: low-temperature regime, intermediate-temperature and high-temperature regime [28]. The low-temperature regime defined as low temperature heat release (LTHR) typically occurs at temperatures from 600 to 800 K [29]. The intermediate-temperature regime is characterized by both the negative temperature coefficient (NTC) zone and the intermediate temperature heat release (ITHR). The high-temperature regime is termed as high temperature heat release (HTHR) and is for temperatures above 1200 K [30]. The main auto-ignition (HTHR) does not occur instantaneously for both single-stage and two-stage combustion [31]. The intermediate temperature heat release (ITHR) is present in both single and two-stage combustion and typically occurs around 950-1050 K [32]. In a study by Curran et al, chemical reactions responsible for the auto-ignition of paraffin's were identified and between the low and high temperature regime, negative temperature coefficient (NTC) region was defined [33, 34]. In another study by Boot et al. the NTC was defined in a temperature range from 850 to 1200 K [30]. This range was selected from the work on chemical kinetics of hydrocarbon auto-ignition performed by Westbrook [35]. Another study defined the ITHR as the heat release between the end of LTHR and the start of HTHR and did not highlight the NTC region [36]. It has been highlighted by Vuilleumier et al. that ITHR is difficult to define and quantify and

further to this Trudesson et al. also highlighted that with an increased engine speed, early shift of the temperature range made the ITHR coincide with the LTHR and NTC region [36, 37]. This was the case when the temperature range was used as an indicator for the characterization of ITHR. In short, it is difficult to come up with general definitions for these heat release rates (ITHR or NTC). As far as this study is concerned, the characteristics described in Figure 9 will be followed for the rest of the paper.

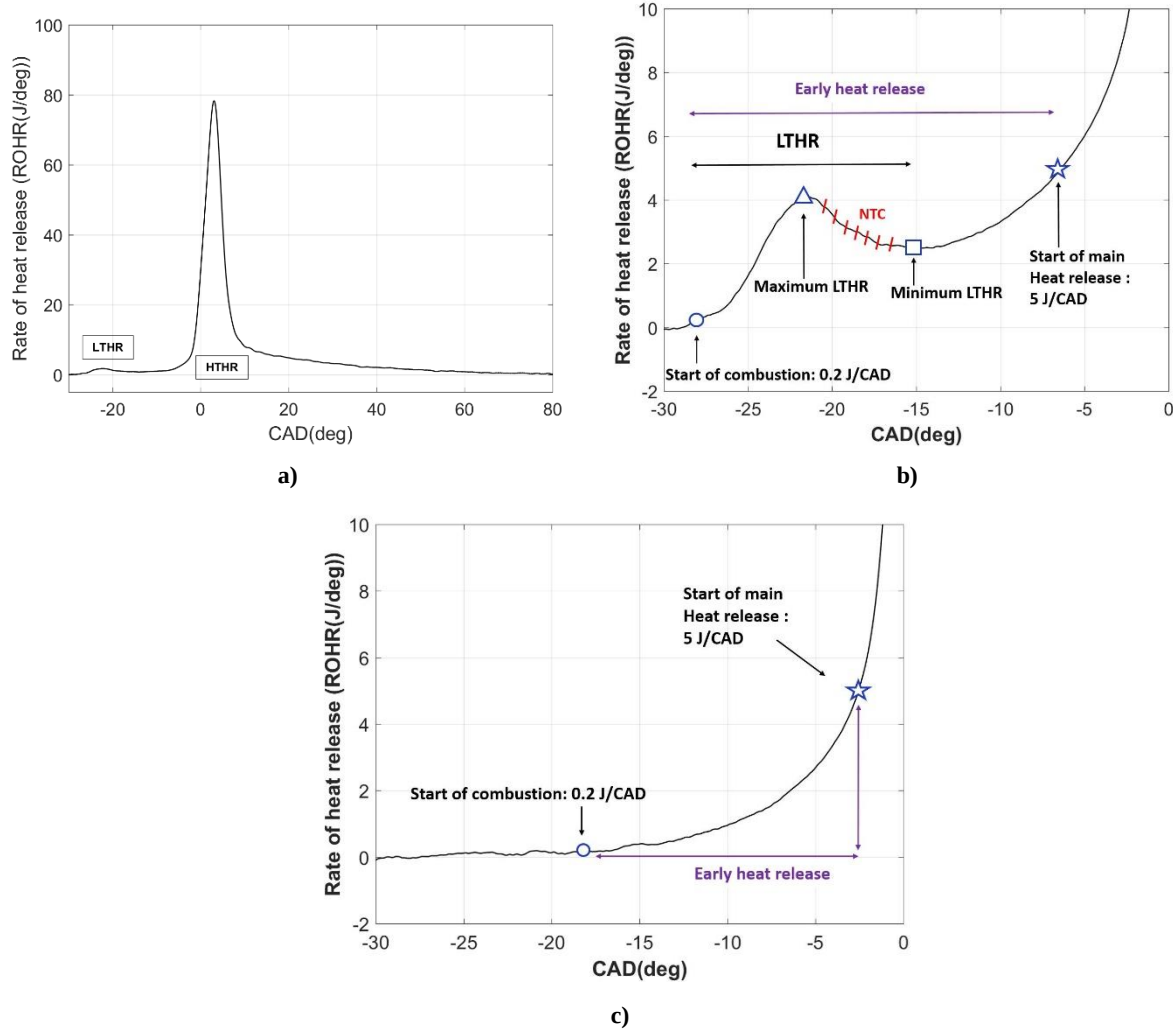


Figure 9 (a, b and c): Definitions of pre-reactions (LTHR, ITHR and NTC) and main heat release (HTHR)

In this study, two definitions have been used for the exothermic pre-reactions or early heat release occurring before the main heat release (HTHR). Figure 9a shows the rate of heat release for one of the fuel blends tested in this study with regions of LTHR and HTHR. Figure 9 b, c illustrates the two definitions for the early heat release. Figure 9b shows the part of early heat release comprising Low temperature heat release (LTHR) and negative temperature coefficient (NTC) with the red marked region showing the reduction in the rate of heat release. In contrast, Figure 9c shows the early heat release without any LTHR and NTC. The start of combustion (open circle) is defined as the rising point of LTHR bump with a threshold of 0.2 J/CAD. This value was selected to avoid any noise effect which would lead to a wrong phasing for the start of combustion. The region of LTHR was defined from the start of combustion to the minimum LTHR (square): the ending of LTHR bump. A threshold value of 5 J/CAD was defined for the main heat release to capture a well-established main heat release (HTHR) and to keep it separate from the intermediate-temperature regime.

The above heat release analysis was used to characterize the auto-ignition behavior of the fuel blends under various operating conditions. The following section will individually illustrate the heat release analysis for PRF 90, Toluene/heptane and Ethanol/heptane. Not all the data points will be illustrated for the heat release analysis in the paper, but a selection of the data points are such that only the main story behind the HCCI testing is highlighted.

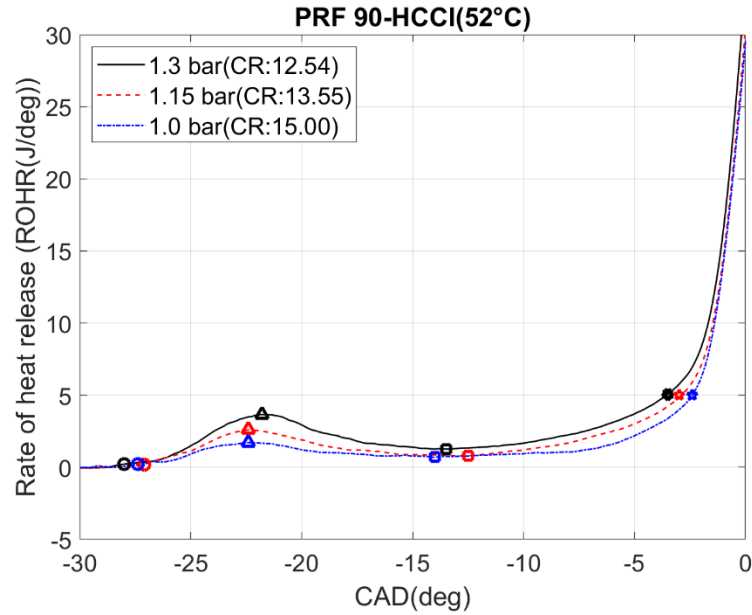


Figure 10: Heat release rate for PRF 90 at 52±5 °C

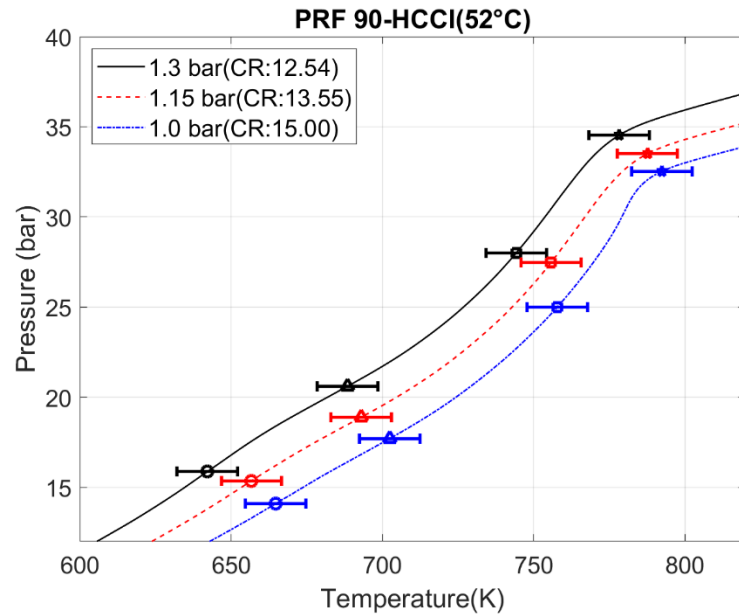


Figure 11: In-cylinder LTHR Pressure-Temperature trajectories for PRF 90 at 52±5°C

Figure 10 shows the low temperature heat release (LTHR) for PRF 90 at different boost pressures. The inlet air-fuel mixture temperature at this condition was 52°C. Figure 11 shows the in-cylinder pressure-temperature trajectories for LTHR shown in

Figure 10. Four different points are highlighted on Figure 10 and 11 showing the start of combustion (circle) which is defined as the rising point of LTHR bump (0.2 J/CAD), maximum LTHR (triangle): maximum value of LTHR bump, minimum LTHR (square) : the ending of LTHR bump and the start of main heat release (star): 5J/CAD. It could be observed that for the highest boost pressure (1.3 bar), the maximum LTHR was found. With a decrease in the pressure, the LTHR decreased. It was not possible to capture the data point for 0.9 bar due to a compression ratio limitation from the hardware. At 1.0 bar, the compression ratio was 15:1 and a further decrease in the intake pressure would have required a higher compression ratio to achieve a combustion phasing of CA50 3° after TDC. The auto-ignition temperature for the maximum LTHR and the start of the main heat release was found to be approximately in the range of 680-700 K and 770-780 K for all the inlet pressures respectively. The uncertainty associated with the estimation of in-cylinder temperature is also shown in the Figure 11. The CAD range for the occurrence of LTHR was from -27 to -13. In addition to LTHR, NTC behavior as defined in Figure 9b was observed for all the three intake pressures. The maximum LTHR for 1.3 bar was estimated to occur at 3.66 J/CAD and this decreased by 29% for 1.15 bar. The change in the compression between 1.3 bar to 1.15 bar was 1.01. With respect to 1.15 bar, the maximum LTHR reduced by 33 % for 1.0 bar whereas the change in the compression ratio was 1.45 which was larger than the change observed at the higher intake pressures (1.3 to 1.15 bar). Overall it is observed that there exists a non-linear behavior between the compression ratio and the maximum LTHR.

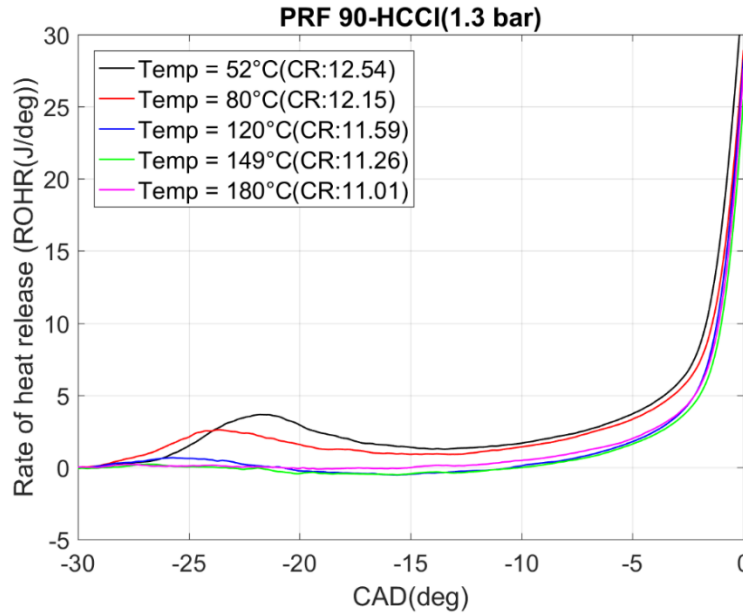


Figure 12: Heat release rate for PRF 90 at different intake temperatures

Further to the intake pressure effect, the effect of intake air/fuel mixture temperature on LTHR for PRF 90 was also investigated. Figure 12 shows the LTHR behavior for a fixed pressure of 1.3 bar with varying inlet air-fuel temperature. It could be observed that maximum LTHR was observed for lowest intake temperature of 52°C and as the inlet temperature increased, the maximum LTHR decreased. With the increase in the intake temperature, the rate of heat release changed from early heat release comprising LTHR and NTC to no observed LTHR at an intake temperature of 180°C. The maximum LTHR differ significantly by varying the inlet temperature which is consistent with findings in the literature [11, 38, 39]. In summary, maximum LTHR is observed with highest boost pressure and lower intake temperature.

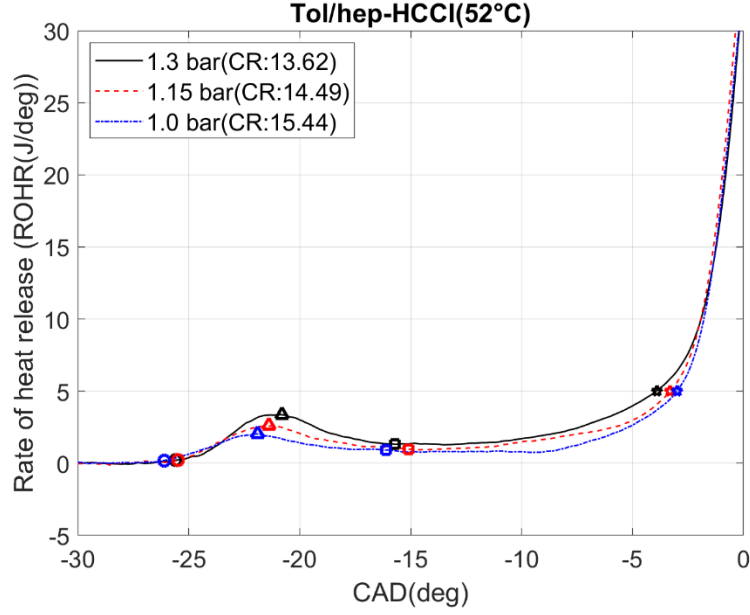


Figure 13: Heat release rate for the toluene/heptane blend at 52±5°C

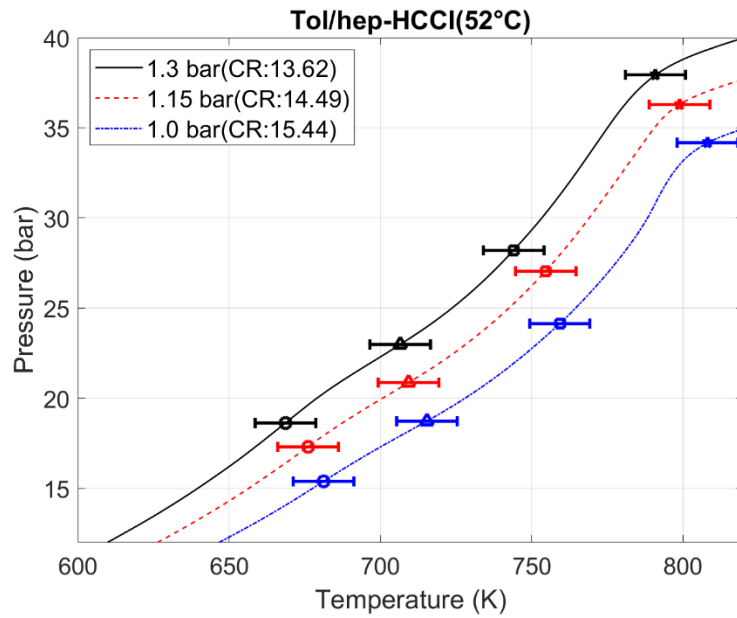


Figure 14: In-cylinder LTHR pressure-temperature trajectories for the toluene/heptane at 52±5°C

In order to explore if the aromatics have any role on the occurrence and magnitude of LTHR, the region of LTHR defined by four points on Figure 9b was further illustrated for the toluene/heptane blend. Figure 13 and 14 shows the LTHR and in cylinder pressure-temperature behavior for the toluene/heptane blend. It can be again observed that with the highest inlet boost pressure of 1.3 bar, maximum LTHR was observed. This was a similar observation when compared to PRF 90. As far as the maximum magnitude of LTHR between the two blends, it was found that PRF 90 had the highest LTHR compared to the toluene/heptane blend at all intake pressures. This highlights that for a given pressure-temperature condition, composition has a role in dedicating the magnitude of LTHR. The CAD range for the occurrence of LTHR for the toluene/heptane blend was about the same range (-

24 to -15 CAD) as observed for PRF 90. The auto-ignition temperature at maximum LTHR was about 720 K and the start of the main heat release was in the range of 780-800 K for all pressure conditions. The maximum LTHR for 1.3 bar was estimated to occur at 3.37 J/CAD and decreased by 23% for 1.15 bar. The change in the compression ratio between 1.3 bar to 1.15 bar was 0.87. With respect to 1.15 bar, the maximum LTHR reduced by 21 % for 1.0 bar whereas the change in the compression ratio was 0.95. This again highlights that the non-linear behavior between the compression ratio and the maximum LTHR exists but the degree of the non-linearity depends on the fuel composition.

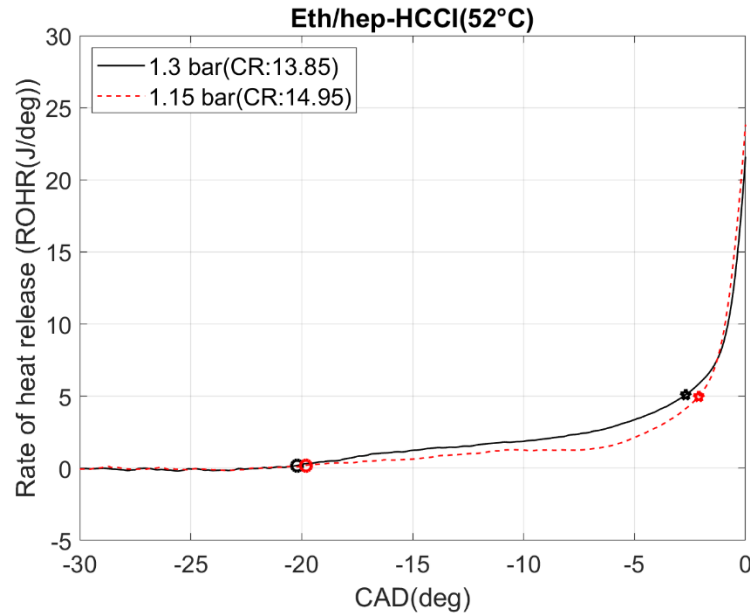


Figure 15: Heat release rate for the ethanol/heptane blend at $52 \pm 5^\circ\text{C}$

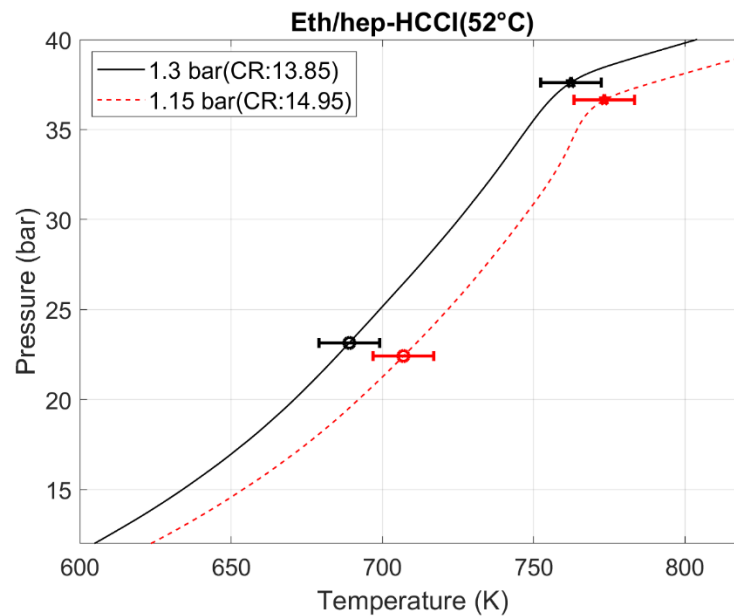


Figure 16: In-cylinder pressure-temperature trajectories for the ethanol/heptane blend at $52 \pm 5^\circ\text{C}$

In order to further explore the compositional effect, in addition to PRF 90 and toluene/heptane blend, heat release analysis were also performed on the ethanol/heptane blends. Figure 15 and 16 show the heat release rate and the pressure-temperature trajectories for the ethanol/heptane blend. The LTHR at an intake pressure of 1.0 bar was not possible to obtain since the CA 50 could not be controlled due to the compression ratio limitation. With the ethanol/heptane blend, it could be observed that no LTHR was detected. The heat release rate included only single stage combustion with only start of combustion and start of main heat release shown similar to the definition of early heat release defined in Figure 9c. The auto-ignition temperature at the start of combustion was found to be higher and the start of the main heat release was lower for the ethanol/heptane blend when compared to the two blends. In summary, for all the three blends, it was observed that maximum LTHR was encountered for boosted conditions and lower intake temperatures.

Discussion

The use of RON and MON tests to describe the anti-knock quality of the fuel is subject to many open questions. The RON and MON are represented by two different trajectories as described in the introduction. Recently Kalghatgi proposed an Octane Index (OI) to better define the anti-knock quality of the fuel in modern engines [3]. One important parameter in the description of octane index (OI) was the numerical value of K [3]. The value of K is 0 in RON whereas in MON test K is 1. Historically the value of K was positive and in 1930 it was 1 but recent tests with modern engines have shown that the value of K is negative [6, 8, 40, 41]. The decreasing trend in the value of K is associated with the better engine cooling, engine breathing and the use of fuel injectors instead of carburetors [8]. This clearly indicates that the modern engines are moving away from both RON and MON trajectories and changes have to be applied to the conventional RON and MON tests to be more applicable to evolving modern engines. Another disadvantage of using the RON and MON test is that the commercial gasoline does not have zero sensitivity ($S = \text{RON} - \text{MON}$) [3]. The scale of RON and MON are based on two paraffinic fuels: iso-octane and n-heptane, and the mixture of the two fuels is termed as Primary Reference Fuels (PRFs) [3]. The sensitivity of the PRF fuels is zero and they do not represent most of the commercial fuels to be used as reference fuels [42]. In addition, recently there has been growing interest in trying to understand the knocking behavior of fuels using Pressure-Temperature (P-T) trajectories [9, 43, 44]. It has been shown that the low temperature reactions before flame propagation are critical in understanding the knocking behavior of engines [9, 44]. The conventional RON and MON tests fail to measure the low temperature reactions due to extensive heat from the propagating flame which again questions the use of these test protocols for octane rating. Despite all the concerns, the RON and MON tests are used for more than 75 years and the current octane rating for modern engines depends on these two tests [42].

As an alternative option, this research paper explores the use of Homogeneous Charge Compression Ignition (HCCI) combustion in the standard CFR engine. Three fuel blends: PRF 90, toluene/heptane and ethanol/heptane blends were tested. It was found that toluene/heptane and ethanol/heptane blends behaved differently to PRF 90 in HCCI mode. The operating conditions such as intake pressure and temperature also played an important role in the occurrence of the LTHR. In reference to intake pressure, other studies from Vuilleumier et al. and Mehl et al. highlighted the sensitivity of LTHR with intake pressure [11, 45] and more specifically Vuilleumier et al operated the engine under HCCI mode and partially throttled up to 1.1 bar with a CR of 12:1 and it was observed that maximum LTHR occurred for the highest intake pressure [11]. The observations from the literature are consistent with the results shown in Figure 10&13 (heat release rate for PRF 90 and Toluene/heptane blends).

As far as the effect of intake temperature on LTHR, Yang et al investigated the role of intake temperature on HCCI combustion using PRF 73 and found that at low intake temperature of 60°C, LTHR was dominant but as the temperature increased to 174°C, LTHR diminished [38]. In another HCCI study by Yang et al, the potential of using boosted HCCI combustion to achieve high loads was investigated and it was again shown that for a constant pressure of 200 kPa, as the intake temperature was increased, LTHR was suppressed [39]. In a HCCI study about primary reference fuels (PRFs) and oxygenated fuels by Trudesson et al, the

reduction of LTHR was observed with an increase in the intake temperature and the rate of heat release translating from LTHR and NTC region at low intake temperature to ITHR at high temperatures [46, 47]. The effect of intake temperature on LTHR for PRF 90 in Figure 12 showed similar observation as highlighted in the literature.

The use of toluene/heptane and ethanol/heptane blends also helped to investigate the compositional effect on LTHR. In the literature, Shibata et al. experimentally investigated the role of fuel composition with LTHR using a supercharged, 4-cylinder high compression engine [48]. It was found that in the presence of aromatics such as toluene, the radicals responsible for the LTHR production was small leading to lower LTHR [48]. Comparatively, the LTHR was quite active for n-paraffins which again highlights that our observations about toluene/heptane (Figure 13) and PRF 90 (Figure 10) blends are consistent with the literature [48]. The paraffinic and aromatic fuels showed the presence of LTHR whereas for ethanol, negligible LTHR was observed as shown in Figure 15. Other studies have also shown that in the presence of ethanol, the LTHR is reduced and for some operating conditions, no LTHR is observed [49, 50]. Ethanol acts as a radical scavenger and its addition to the base fuel significantly reduces the LTHR [50]. The inlet air temperature in Figure 15 was 52°C and the other studies by Waqas et al. and Singh et al. have shown the presence of LTHR at this temperature but in our study no LTHR was found [49, 50]. In these studies, the maximum amount of ethanol added to the base fuel was 10% and 20% by volume and the experimental data was collected at 1 bar [49, 50]. Vuilleumier et al [51] conducted experiments with ethanol/heptane blends and found that with 40% heptane/60 % ethanol, LTHR was observed. The experiments were conducted with the boosted pressure of 2.2 bar [51]. The possible explanation that no LTHR was observed in our experiments (Figure 15) was due to both fuel composition and the operation conditions.

The research work shown in this paper was written with the purpose to highlight the significance of HCCI combustion using an unmodified CFR engine. With the HCCI octane rating method already existing [24], the use of HCCI combustion could provide an alternative path to rate the fuels in the standard CFR engine. This will require more experiments to be conducted to highlight the similarity in the auto-ignition behavior for HCCI with SI end gas auto-ignition. The transfer function used in this study was based on a paraffinic scale similar to the current scale used in RON and MON tests. In order to take sensitivity (S) into account, it is important that future experiments are based on the fuels have sensitivity. One option could be the use of n-heptane and toluene as suggested by Kalghatgi [19].

Overall this work provides initial insight into the use of HCCI combustion in the CFR engine. Further work is required which includes the use of toluene/n-heptane as octane rating scale, collection of more HCCI data by testing range of different fuels and uncertainties associated with the Lund-Chevron HCCI fuel number needs to be defined before this combustion mode can be officially adopted for the octane rating in SI engines.

Conclusions

The use of HCCI combustion with the standard CFR engine was demonstrated for the first time as a proposal for rating the fuels in SI engines. Following are the conclusions obtained from the trends: -

1. The successful operation of HCCI combustion in the standard CFR engine demonstrates that other octane rating methods could be adopted given that there is a lot of uncertainty surrounding RON and MON.
2. The use of HCCI combustion along with the standard CFR engine demonstrates that it is possible to detect LTHR in the standard CFR engine and could possibly help to understand the knocking behavior of the fuels.
3. The ability to run the CFR engine with various operating conditions (intake pressures/temperatures) shows that the standard CFR engine can replicate modern engine operating conditions which do not represent RON and MON conditions.
4. Low temperature reactions were found to be a function of intake pressure, temperature and composition of the fuel with maximum LTHR being observed under the boosted pressure conditions.

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Appendix A

Heat release analysis [26, 50, 52-54]:

For the three blends, pressure data for 300 cycles were used for the heat release analysis. A pressure filtering process is applied to the data to make the traces smooth. Both the motored and firing traces were used for the analysis. The first law heat release analysis is used to derive the equation A shown below: -

First law heat release can be given by:

$$\frac{dQ}{d\theta}_{Apparent} = \frac{\gamma}{\gamma - 1} \frac{PdV}{d\theta} + \frac{1}{\gamma - 1} \frac{VdP}{d\theta} \quad (A)$$

Equation A refers to the apparent heat release. A Matlab code is written which helps to calculate the rate of apparent heat release.

The ratio of specific heat (γ) is estimated using a linear relation and is given by :-

$$\gamma = \gamma_o - \frac{T_{cylinder} - 300}{1000} \cdot k \quad (B)$$

Following are the values used for both motored and combustion cases: -

	γ_o	k
Motored	1.40	0.07
Combustion	1.37	0.05

The heat release was tuned for both motored and combustion pressure traces. For the motored trace, the tuning was performed to make sure that the heat release is close to zero as much as possible.

Heat transfer to the walls is based on the Woschini analysis. It is given by:

$$h = 131 \times (Bore)^{-0.2} \times Incylinder \text{ Pressure}^{0.8} \times Incylinder \text{ Temperature}^{-0.55} w^{0.8}$$

Where

$$w \text{ (Charateristic gas velocity)} = C_1 \times 2.28 \times \text{Mean piston speed} + C_2 \times 3.24 \times 10^{-3} \times \frac{V_{displaced}}{V_{IVC}} \times \left(\frac{P_{fir} - P_{mot}}{P_{IVC}} \right) \times T_{IVC}$$

Where C_1 and C_2 were constants and tuned such that the heat release was close to zero for the motoring cases.

V_{IVC} , P_{IVC} and T_{IVC} = volume, pressure and temperature in the cylinder when the inlet valve closes. T_{IVC} was assumed to be equal to inlet air temperature. P_{IVC} was obtained from the experimental pressure data.

P_{fir} and P_{mot} = Burnt pressure and motored pressure trace

The heat release loss is given by:

$$\frac{dQ}{d\theta}_{ht} = h \times Area \times (T_{Incylinder} - T_{wall})$$

The temperature of the walls was assumed to be equal to the coolant temperature and was assumed to be 370 K.

$$T_{Incylinder} = \frac{Pressure_{Incylinder} \times Volume_{Incylinder} \times Temperature_{IVC}}{Pressure_{IVC} \times Volume_{IVC}}$$

The net heat released in a cycle is the sum of the considered values:

$$\frac{dQ}{d\theta} = \frac{dQ}{d\theta}_{Apparent} + \frac{dQ}{d\theta}_{heat\ loss}$$

The crevices and blow by losses were assumed to be negligible in this calculations.

Rebuttal JENG-2019-0040

As instructed, following rebuttal has been prepared to address the comments from the reviewers. Each reviewer has been assigned a color and the changes are highlighted in the main text. The reviewer's comments are in bold.

Reviewers' comments:

Editor:

Editor: Please consider revising the manuscript title to reflect the main goal and content of this work - characterization of low-temperature reactions

As instructed, the title has been changed. For your quick reference, the title is given below:-

Characterization of low temperature reactions in the standard Cooperative Fuel Research (CFR) Engine. (Page 1)

Reviewer #1:

The experimental results are interesting. However, the reviewer recommends more discussion. Please mention on that high compression ratio causes low LT HR with high compression temperature while the relationships in Figs. 6-8 between T and CR show different tendency, depending on fuel.

As instructed, the author has included more discussions in the result section and is highlighted in yellow in the main text. Also in the discussion section, the results were compared with the other studies in the literature review. (Page 13-17)

Regarding the question about Figure 6-8 and the relationship between T, CR and LTHR, the following comments are provided by the author: -

The HCCI combustion is mainly driven by auto-ignition of the fuel and the two main parameters, compression ratio and intake temperature are used to control the combustion. For higher compression ratios, lower intake temperatures are required and vice versa.

In order to capture the data point, a criterion of CA 50 of 3°ATDC was used. For a given intake temperature, the compression ratio was varied until the criterion is achieved. For the low intake pressure, higher compression ratio was required to achieve CA 50 of 3°ATDC and hence low LTHR was observed.

Reviewer #2:

This study investigated the knocking behavior with HCCI combustion in the standard CFR engine. The results from the study would be helpful to understand the auto-ignition behavior of various fuels. This study is well suited to the purpose of the SAE international journal. Therefore, the reviewer recommends minor revision for this paper.

Issue 1

Page 1, Introduction

The lumping references [1-7] should be avoided. Instead, the main finding or contribution of each study should be summarized in separate sentences.

The first sentence of the paper was more general in nature and it is well known fact among the engine research community. Therefore the references to this sentence has been removed. Followed by the first line, the author has tried to explain each reference separately and where appropriate, multiple references has been avoided.

Issue 2

Page 2, Introduction

It would be helpful to add other studies that investigated the auto-ignition characteristic based on HCCI combustion.

Changes applied and are highlighted in green color in the introduction section. (Page 3)

Issue 3

Page 4, Test Fuels and operating conditions

Please use the abbreviation, because the full name of the CFR was declared previously.

Changes applied as requested. The required text has been changed to ‘The standard CFR engine was operated under both stoichiometric and lean conditions to obtain the Auto-ignition Metrix for the rating of the fuels’.(Page 5)

Issue 4

Page 4, Test Fuels and operating conditions

The reason of selecting the excess air ratio of three in the HCCI tests should be presented in the manuscript.

Changes applied as requested. The following sentence has been added in the main text: - A diluted charge is required in HCCI combustion to keep the pressure rise rate below an acceptable level. (Page 5)

Issue 5

Page 4, Test Fuels and operating conditions

How many times each experiment was repeated and what is the COV of the measured parameters?

Changes applied as requested. The author believes that the reviewer was referring to the compression ratio measurements in Figure 6 to 8. The following explanation has been added in the main text to address the comments from the reviewer. Also uncertainty bars have been included on the Figure 6 to 8: -

‘In addition to the above observations, it was realized during the testing that small changes in the compression ratio can lead to large changes in the slopes for Figure 6-8. Therefore, a daily test was carried out on the PRF 90 blend at the start and the end of the experiments to check the consistency of the obtained compression ratios. On average, the uncertainty percentage for PRF 90 was found to be 2%. Figure 6-8 also include the uncertainty bars for the plotted data based on the daily test of PRF 90. These fall within the marker size’. (Page 8-10)

In addition to the above comments in the main text, the standard ‘Engine Fit for Use’ test according to ASTM method was also conducted every 3 days.

Issue 6

Page 6, Results

**The MON data was obtained from the word done by
-> The MON data was obtained from the work done by**

Correction applied in the main text. (Page 7)

Reviewer #3:

Fig. 1 is not easy to understand with the given description. Citation of Literature 13 seems to be wrong, actual paper number in SAE database is 2005-01-2083.

Fig. 10 to 16: if they are not coloured it is not possible to distinguish the lines
check spelling, sometimes "was" instead of "were" is used in plural

As instructed, Figure 1 has been modified and some more description has been included on the Figure 1. (Page 2)

The author is thankful to the reviewer for identifying an error in the reference. The Reference has now been corrected.

Spelling and grammar checks applied.

The author has also improved the figures 10-16 except from Figure 12, where due to large number of lines, it was difficult to come up with different line styles. In addition, the author has also added some uncertainty bars for the Pressure-Temperature (P-T) plots of the three fuel blends (Page 12-15)